Research Quarterly for Exercise and Sport

ISSN: 0270-1367 (Print) 2168-3824 (Online) Journal homepage: www.tandfonline.com/journals/urqe20

Effect of Strength Training on Rate of Force Development in Older Women

André Luiz Demantova Gurjão, Lilian Teresa Bucken Gobbi, Nelson Hilário Carneiro, Raquel Gonçalves, Rodrigo Ferreira de Moura, Edilson Serpeloni Cyrino, Leandro Ricardo Altimari & Sebastião Gobbi

To cite this article: André Luiz Demantova Gurjão, Lilian Teresa Bucken Gobbi, Nelson Hilário Carneiro, Raquel Gonçalves, Rodrigo Ferreira de Moura, Edilson Serpeloni Cyrino, Leandro Ricardo Altimari & Sebastião Gobbi (2012) Effect of Strength Training on Rate of Force Development in Older Women, Research Quarterly for Exercise and Sport, 83:2, 268-275, DOI: 10.1080/02701367.2012.10599857

To link to this article: https://doi.org/10.1080/02701367.2012.10599857

Published online: 23 Jan 2013.

Submit your article to this journal

Article views: 562

View related articles

Citing articles: 4 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=urqe20

Research Quarterly for Exercise and Sport
©2012 by the American Alliance for Health,
Physical Education, Recreation and Dance
Vol. 83, No. 2, pp. 268–275

Effect of Strength Training on Rate of Force Development in Older Women

André Luiz Demantova Gurjão, Lilian Teresa Bucken Gobbi, Nelson Hilário Carneiro, Raquel Gonçalves, Rodrigo Ferreira de Moura, Edilson Serpeloni Cyrino, Leandro Ricardo Altimari, and Sebastião Gobbi

We analyzed the effect of an 8-week strength training (ST) program on the rate of force development (RFD) and electromyographic activity (EMG) in older women. Seventeen women (M age = 63.4 years, SD = 4.9) without previous ST experience were randomly assigned to either a control (n = 7) or training (n = 10) group. A leg-press isometric test was used for assessment. ST (three sessions/week, three sets of 10–12 repetition maximum, five different exercises) induced significant increases (p < .05) on peak RFD (48.4%) and on RFD and EMG of vastus medialis at time intervals of 0–50, 0–100, 0–150, and 0–200 ms (41.1–69.2% and 43.8–64.3%, respectively). Therefore, ST resulted in favorable changes in neuromuscular responses in older women.

Key words: aging, muscle activation, neuromuscular performance, resistance training

Significant alterations in the neuromuscular system accompany aging. Progressive loss of strength and power contributes to disability and loss of independence and quality of life in older adults. This is particularly applicable to women, because they have a longer life expectancy, reduced functional reserve, and a greater decline in muscular strength in the lower limbs than do men (Kwon, Schrager, Talbot, Fozard, & Metter, 2001; Lynch et al., 1999; Skelton, Kennedy, & Rutherford, 2002). Previous studies suggested that age-related declines in muscle power occur at a greater rate (3–4% per year) than for strength (1–2% per year; Izquierdo et al., 1999; Skelton, Greig, Davies, & Young, 1994). Foldvari et al. (2000) reported that muscle power might be of greater importance than muscle strength for tasks such as rising from a chair and climbing stairs. In addition, maintaining an adequate level of muscle power could aid in fall prevention in older adults (Pijnappels, van der Burg, Reeves, & van Dieën, 2008).

In this context, strength training (ST) has been shown to be effective in improving neuromuscular performance in older adults. Indeed, in a systematic review, Latham, Bennett, Stretton, and Anderson (2004) demonstrated that ST can safely increase muscle mass and power in older adults and improve physical function. In addition, other studies that investigated the effects of ST on muscle power described changes in jump performance (Kalapotharakos, Smilios, Parlavatzas, & Tokmakidis, 2007) or maximal muscle power, calculated from torque-angular velocity curves (Andersen et al., 2005). However, few studies have investigated the effect of ST on the rate of force development (RFD) in older women, measured in the initial phases of muscle action.

Submitted: June 30, 2010
Accepted: March 25, 2011

André Luiz Demantova Gurjão; Raquel Gonçalves and Sebastião Gobbi are with the Laboratory of Physical Activity and Aging at UNESP—Universidad Estadual Paulista. Lilian Teresa Bucken Gobbi is with the Gait and Posture Studies Laboratory at UNESP—Universidad Estadual Paulista. Nelson Hilário Carneiro is with the Research and Study Group in Exercise and Aging at UNOESTE—Universidad Oeste Paulista. Rodrigo Ferreira de Moura is with the Laboratory of Cell Signaling at UNICAMP—Universidad Estadual de Campinas. Edilson Serpeloni Cyrino and Leandro Ricardo Altimari are with the Research and Study Group in Metabolism, Nutrition and Exercise at UEL—Universidad Estadual de Londrina.

RFD, which is defined as the ability to produce muscle force rapidly, determines the magnitude of acceleration in the initial movement phase and, ultimately, influences the movement velocity (Aagaard, Simonsen, Andersen, Magnusson, & Dyhre-Poulsen, 2002; Caserotti, Aagaard, Larsen, & Puggaard, 2008; Kraemer & Newton, 2000). In addition, RFD plays a key role in the development of maximal muscle power and has been reported to correlate with capacity to maintain postural control (Aagaard et al., 2002; Kraemer & Newton, 2000). During an attempt to prevent a fall, an individual needs to develop force in a short period of time (0–200 ms). Therefore, in such a context, RFD plays a more important role than maximal voluntary contraction (MVC; ~ 400–600 ms; Aagaard et al., 2002; Suetta et al., 2004). In older adult fallers, for example, RFD was significantly lower compared to nonfallers (Pijnapples et al., 2008).

Several previous studies investigated the effect of ST on RFD in older adults. Suetta et al. (2004), for example, examined the effect of three training regimes (ST, electrical muscle stimulation, and standard rehabilitation) on RFD after unilateral prolonged disuse due to osteoarthritis. Only ST induced significant increases (26–45%) in RFD, accompanied by significant increases in a cross-sectional area of the quadriceps muscles. Significant differences in electromyogram (EMG) amplitudes were observed with 5–12 weeks of ST after hip-replacement surgery. Barry, Warman, and Carson (2005) investigated differences in rapid muscle activation of the elbow flexors after ST between young and older adults. For both groups, the increase in RFD was accompanied by an increase in the EMG amplitude throughout the initial 100 ms of activation. For older adults, however, this training response was apparent only in the monofunctional muscles (brachialis and brachioradialis). The results of both studies are important for understanding the effects of ST on RFD, but such findings cannot be applied to nonclinical populations or lower-limb performance. Although Caserotti et al. (2008) showed a positive effect of ST on RFD in older adults (increases from 21% to 51%), they investigated changes only for the 0–200-ms time interval relative to the onset of muscular contraction and did not assess muscle activation. In young adults, it has been shown that improved RFD following resistance training is accompanied by an increase in the surface EMG (Barry et al., 2005; Van Cutsem, Duchateau, & Hainaut, 1998). Because loss of strength can affect independence in older women, it is important to determine whether they experience such adaptations in the initial phase of muscle activation, following ST. We hypothesized that following training there would be an increase in (a) peak RFD, (b) RFD at intervals of 0–50, 0–100, 0–150, and 0–200 ms (relative to the onset of muscular contraction), and (c) EMG amplitudes recorded for the same RFD time intervals. Therefore, the purpose of this study was to investigate the effects of an 8-week strength training program on RFD and muscle activation in older women.

Method

Participants

Twenty-one older women from the local community were randomly assigned to two groups: training (TG; n = 11) and control (CG; n = 10). Exclusion criteria for participation in the study were the presence of uncontrolled hypertension or participation in any systematic physical exercise (e.g., strength, stretching, balance, or aerobic training) in the 3 months prior to beginning the study. A physician examined participants and reported no known neurological or musculoskeletal disorders. Participants were informed verbally about the procedures they would undergo, and each signed an informed consent form. The local Institutional Committee on Ethics in Research approved this study. One participant in the TG and 3 in the CG dropped out due to personal reasons or orthopedic problems unrelated to the intervention. Thus, 10 participants in the TG (M age = 61.7 years, SD = 4.8; M height 157.3 cm, SD = 7.7; M weight 62.1 kg, SD = 12.4), and 7 in the CG (M age = 65.0 years, SD = 5.1; M height 156.6 cm, SD = 5.3; M weight 63.6 kg, SD = 11.2) completed the study.

Procedures

Measurement of Isometric Force-Time Curve and Processing. In both groups, RFD and MVC were assessed before and after the experimental period. Before the assessments, one practice session was conducted to familiarize participants with data recording procedures for the isometric force-time curve (Cf-t). The bilateral isometric Cf-t of the knee and hip extension were assessed using a device consisting of a metal frame, as described by Sahaly, Vandewalle, Driss, and Monod (2001), which enabled adjustment of the sliding seat knee angle (90º). We chose this angle for both pre- and posttraining, because some activities of daily life, no matter the training status, require muscle force output at different angles compared to optimum ones (e.g., standing up from a chair or climbing stairs). In addition, during the balance recovery phase, for example, it is possible that high knee and ankle extension force output are also performed at nonoptimal angles. The hip joint angle was fixed at 110º. The leg press consisted of a foot platform fixed to a frame that made contact with a force transducer (model 2000 N™, EMG system®, São José dos Campos, SP, Brazil). The hip was fixed, and participants were allowed to stabilize their upper body by holding on to handles attached to the leg press. Each participant's position was recorded to ensure it was identical under pre- and postconditions. Immediately before each assessment,

participants were instructed to produce an MVC, "as fast as possible," for 5 s. As soon as participants began the effort, they were verbally encouraged to exert a maximum effort and received visual feedback regarding their performance. Each participant performed the test three times, with a recovery interval of 3–5 min between each trial.

Signal acquisition from the force transducer was made using an analog signal amplifier (model CS 800 AF™, EMG system®, São José dos Campos, SP, Brazil), at a sample frequency of 2000 Hz, and synchronized to all the EMG signal recordings. The obtained signal was stored on a hard disk and analyzed off-line afterward. First, using a cutoff frequency of 15 Hz, a fourth-order, zero-lag Butterworth low-pass filter was used to digitally filter the raw signal from the force transducer. We defined the onset of contraction as the point at which the muscular force value exceeded 7.5 N above the baseline (Aagaard et al., 2002). The MVC was determined as the highest value registered within the 1-s window (500–1,500 ms) that corresponded to the onset of muscular contraction in each trial (Häkkinen et al., 2001). The peak RFD (PRFD) was determined by the steepest slope of the curve, calculated within regular windows of 20 ms (ΔForce/ΔTime), for the first 200 ms after contraction onset. RFD values were obtained at time intervals of 0–50; 0–100; 0–150; and 0–200 ms, relative to the onset of muscular contraction (Aagaard et al., 2002).

Electromyography Recording and Processing. Biological signals were recorded according to the ISEK (Merletti, 1999), using an 8-channel module (model CS 800 AF™, EMG system®, São José dos Campos, SP, Brazil), with a band pass filter with cutoff frequencies at 20–500 Hz, an amplifier gain of 1,000X, and a common mode rejection ratio > 120 dB. A converting plate for A/D 12–bit signal was used to convert the analog to digital signals, with a sampling frequency of 2000 Hz for each channel and an input range of 5 mV (de Andrade et al., 2005).

Electrode placement on anatomical points followed standards proposed by the Surface Electromyography for the Non–Invasive Assessment of Muscles (Hermens, Freriks, Disselhorst-Klug, & Rau, 2000). Bipolar surface electrodes (diameter: 10 mm; center-to-center distance: 23 mm) were placed over the agonist vastus medialis (VM) and vastus lateralis (VL) of the right leg. The longitudinal axes of the electrodes were in line with the presumed direction of the underlying muscle fibers. All electrode positions were carefully measured to ensure identical pre- and posttraining recording sites. To lower skin impedance, locations for electrode placement were carefully prepared (shaved, then cleaned with an abrasive and alcohol), and a layer of electrolytic gel was applied to the electrode surface.

A high-pass, fourth-order, zero-lag Butterworth filter with a 5-Hz cutoff frequency was used to digitally filter the EMG signal, followed by a moving root mean square filter with a time constant of 50 ms (Aagaard et al., 2002). We determined the integrated electromyography (iEMG) activity to be the area below the curve of the envelope signal, for the different isometric force-time curve parameters. Thus, iEMG activity was in time intervals of 0–50, 0–100, 0–150, and 0–200 ms, relative to the onset of iEMG, which was initiated 70 ms before onset of the contraction to account for an electromechanical delay (Aagaard et al., 2002). The iEMG activity for the MVC was determined within the 1-s window (500–1,500 ms) corresponding with the onset of muscular contraction (Häkkinen et al., 2001).

Strength Training Program. Participants engaged in a fully supervised, total body ST program for 3 days per week, on nonconsecutive days, for a period of 8 weeks. Although the focus of the intervention was the lower extremity, participants performed other exercises to provide an overall conditioning stimulus. The training program consisted of five exercises performed with three sets of 10–12 repetitions maximum (RM), performed in the following order: chest press; leg press; lat pull down; triceps pulley; and arm curl. Participants performed all exercises bilaterally, with the concentric movement phase lasting ~ 2 s, followed by ~ 3-s eccentric movement phase.

The initial training load was determined during three consecutive sessions, 48 hr apart, which ended 1 week before the experimental protocol began. Participants performed two sets of 12 repetitions and a third set to fatigue. The load was adjusted so that the fatigue would occur at a maximum of 10 repetitions in the last set (Benson, Docherty, & Brandenburg, 2006). Whenever participants exceeded the maximum number of preestablished repetitions (12 RM) in the last set, the load was sufficiently increased to bring the number of repetitions back within the training zone (10 RM). The recovery period between sets and exercises ranged from 60 to 90 s.

Statistical Analysis

To examine the training-related effects on RFD at different time intervals, PRFD, and MVC, we used a 2 x 2 analysis of variance, Groups (TG and CG) x Time (pre and post), with repeated measures on time. To provide a basis on which to conduct comparisons across the groups, we calculated effect sizes (ES). We performed the U Mann-Whitney U and Wilcoxon tests for the iEMG activity obtained during RFD for inter- and intragroups comparison, respectively. We used Pearson correlation coefficients to assess linear relations between absolute changes in iEMG and RFD. We used an intraclass correlation coefficient (R) to test the reliability of MVC and peak RFD. The statistical significance level for all analyses was $p < .05$.

Results

The adherence level for the TG was high, with participants attending 97% of the exercise sessions. Two

participants were absent during two sessions, and 1 was absent during one session. There were no adverse events with ST. The intraclass correlation coefficients for MVC and peak RFD were .94 (95% confidence interval [CI]; .85–.97) and .84 (95% CI; .63–.93), respectively. The MVC showed a significant Group x Time interaction, $F(1, 15) = 9.27$, $p < .05$, ES = .38. The TG increased the MVC by 19% (from $M = 664.3$ N, $SD = 66.7$, to $M = 787.7$ N, $SD = 66.2$) compared to 2.3% in the CG (from $M = 826.8$ N, $SD = 75.2$, to $M = 846.1$ N, $SD = 84.6$).

Table 1 shows the RFD results. There were significant Group x Time interactions for all RFD analyzed (PRFD: ES = .26; RFD 0–50 ms: ES = .17; RFD 0–100 ms: ES = .29; RFD 0–150 ms: ES = .37; RFD 0–200 ms: ES = .33). There was a main effect for time in all RFD, with except for the first 50 ms. In the TG, increases in RFD (0–50 ms; 0–100 ms; 0–150; and 0–200 ms) were accompanied by an increase in their respective iEMG activity, recorded from the VM after the experimental period (see Figure 1-A). In contrast, iEMG activity recorded from the VL displayed an increase for all time intervals, but there was no significant difference (see Figure 1-B). There were no significant changes in iEMG activity for the CG after the experimental period in both VM and VL (see Figure 1-A and B).

The changes in RFD during the specific time intervals (0–50 ms; 0–100 ms; 0–150; and 0–200 ms) were not correlated significantly with corresponding changes in iEMG activity of VM ($R^2 = .04$, .32, .07, and .01, respectively; $p = .14$–.93) and VL ($R^2 = .04$, .01, .08, and .20, respectively; $p = .19$–.86) or together (VM+VL/2; $R^2 = .01$, .08, .01, and .04, respectively; $p = .50 – .97$).

Discussion

The purpose of this study was to analyze the effect of an 8-week ST program on RFD and muscle activation in older women. The results show that ST increased PRFD, RFD at different time intervals, and MVC and that a concomitant increase in iEMG activity for the VM muscle accompanied these modifications. By using the root mean square filter, it was possible to identify changes in muscle activation in the early phase of rising muscle force (0–50 ms), which was also reported in previous studies involving older adults (Barry et al., 2005; Suetta et al., 2004).

In accordance with the American College of Sports Medicine (ACSM, 2009) position stand, the ST protocol used in our study can be characterized as low-to-moderate volume for the quadriceps muscle (three sets of 10–12 RM), performed at a moderate-to-slow velocity. We observed significant increases in muscle activation and in capacities for developing force rapidly. These alterations suggest that older women without previous ST experience do not need complex training routines to improve RFD and muscle activation. It is well known that participation in ST programs involving rapid force production encour-

Table 1. Results of the rates of force development for different time instants, pre, and post 8- week experimental periods

Variables	TG (n = 10)		CG (n = 7)		Effects	F	p
	M	SE	M	SE			
PRFD (N.s^{-1})							
Pre	2,255	504	3,002	406	Group	0.01	.90
Post	3,346	567	2,952	514	Time	5.41	.03
Δ%	48.4	-1.7			Group x Time	6.83	.02
RFD 50 ms (N.s^{-1})							
Pre	1,541	360	1,986	273	Group	0.37	.55
Post	2,607	521	1,846	413	Time	3.18	.09
Δ%	69.2	-7.0			Group x Time	5.39	.03
RFD 100 ms (N.s^{-1})							
Pre	1,727	388	2,241	334	Group	0.03	.85
Post	2,616	448	2,271	450	Time	6.27	.02
Δ%	51.5	1.3			Group x Time	5.48	.03
RFD 150 ms (N.s^{-1})							
Pre	1,654	333	2,191	308	Group	< .01	.93
Post	2,497	399	2,234	399	Time	8.95	< .01
Δ%	51.0	2.0			Group x Time	7.65	.01
RFD 200 ms (N.s^{-1})							
Pre	1,533	282	2,034	269	Group	0.01	.91
Post	2,168	295	2,073	309	Time	7.51	.01
Δ%	41.4	1.9			Group x Time	6.06	.03

Note. TG = strength training group; CG = control group; M = mean; SE = standard error; RFD = rate of force development; PRFD = peak RFD; Δ% = percent change in relation to preexperimental period.

ages improvements in RFD (Blazevich, Horne, Cannavan, Coleman, & Aagaard, 2008). However, performing resistance exercises without the fast force production component has also been shown to have a similar effect (Henwood, Riek, & Taaffe, 2008). Among young adults, Blazevich et al. (2008) found that ST programs with this characteristic might improve RFD in individuals who lack the ability to attain the prescribed maximum force output levels. Increases in RFD that result from traditional speed training have been shown to occur concurrently with increases in agonist muscle activation, as assessed by surface EMG recordings (Aagaard et al., 2002; Suetta et al., 2004). However, it is important to note that they measured muscle force output at only one joint angle (90º). Reeves, Narici, and Maganaris (2004) reported that ST caused a change in the optimal knee joint angle for torque generation, toward full knee extension (from 70º to 60º, before and after training, respectively). In addition, the increases in torque were not constant across joint angles. In this context, the changes we found might have been even greater if our postintervention measurement had been performed at lower angles.

When performing daily living activities, older adults can be required to perform movements in a shorter time than necessary for them to reach MVC, such as when attempting to avoid a fall (0–200 ms; Suetta et al., 2004). For such conditions, the PRFD and RFD at different time instants assume a more important role than muscular strength itself (Aagaard et al., 2002). In addition, the rapid production of force (0–30 ms) is also associated with the capacity to walk rapidly (Suetta et al., 2004). From a practical perspective, an improvement in older women's capacity to develop force rapidly can have an important impact on their daily living activities and a reduction in their risk of falls (Pijnapples et al., 2008). RFD improvement can decrease total response time by reducing the movement time. However, total response time also involves premotor time, which is slower in older individuals (Godefroy, Rous-

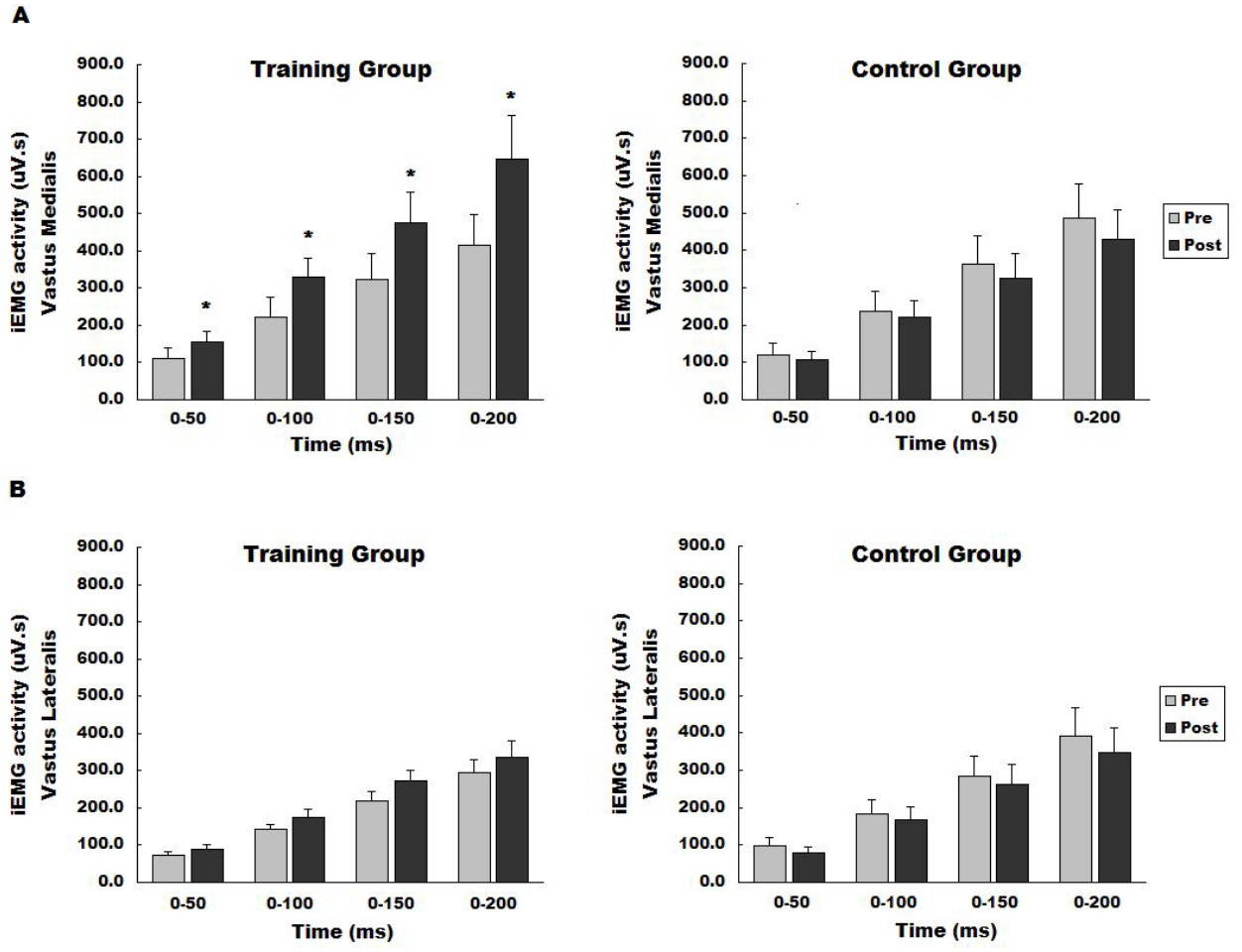

Figure 1. Integrated electromyographic (iEMG) activity for the vastus medialis (A) and lateralis (B) during the rate of force development (RFD) in the different time instants relative to the onset of muscle action (0–50; 0–100; 0–150; 0–200 ms), pre and post 8-week experimental period for strength training (n = 10) and control (n = 7) groups. * Significant differences from preexperimental period (p < .05). Values are means and standard error.

sel, Despretz, Quaglino, & Boucart, 2010) and cannot be assessed by RFD. Furthermore, one study (not involving ST) showed that exercise training improved premotor time (Marmeleira, Godinho, & Fernandes, 2009). With respect to integrating afferent information into movement, it has been reported that older adults present a slower total response time for the lower limbs compared to upper limbs (Li, McColgin, & Van Oteghen, 1998). Thus, an increase in the capacity to develop lower limb force rapidly can be an important training strategy for reducing total response time in older adults. Clinically, ST may be a particularly beneficial training mode for older adults at risk for mobility disability or other strength/muscle related disorders and can lead to enhanced quality of life (Katula, Rejeski & Marsh, 2008).

In the present study, different neural adaptations could have occurred that contributed to the increase in EMG signal amplitude, especially during RFD occurrence. ST could have lead to additional motor unit recruitment during the early training phase and may account for gains in muscular strength and EMG amplitude. In addition, Van Cutsem et al. (1998) showed that motor units are activated earlier after a training period, which may impose a greater effect on RFD compared to MVC. However, improvements in muscular activation after ST have been shown to be more highly correlated with increases in motor unit discharge rate (Kamen, 2005).

Recent studies suggested that an age-related reduction in the PRFD or RFD at different time intervals is due, in great part, to a slowing of the muscle contractile properties and decreases in both the maximal discharge rate and double discharge (Christie & Kamen, 2006; Klass, Baudry & Duchateau, 2008). Double discharges are short interspike intervals (less than 20 ms) in a train of motor unit potentials and are more prevalent at the onset of muscular contraction (Garland & Griffin, 1999). Although their function is not fully understood, it has been suggested that double discharges are particularly useful for achieving high forces or RFD (Garland & Griffin, 1999). A main neural adaptation resulting from ST, and which has a profound impact on RFD, is an increase in the maximal discharge rate of motor units and doublets (Kamen & Knight, 2004; Van Cutsem et al., 1998). Furthermore, Garland and Griffin argued that the impact of the double discharges on whole-muscle force production would depend on the prevalence of their use by motor units. However, our surface EMG recording did not allow us to make a distinction as to which adaptation had a greater or lesser influence.

Blazevich et al. (2008) showed significant correlations between muscle activation parameters (rate of EMG rise and mean average voltage) and RFD in the responders to ST. Our study analyzed only the iEMG activity; the lack of significant correlation between the variation in iEMG activity and RFD may indicate that mechanisms other than neural adaptations had a greater influence on the increase in RFD. It is important to consider the limitations of surface EMG recordings. Thus, another interpretation of our data is that the increase in iEMG activity also could be influenced by peripheral adaptations, such as alterations in the thickness of subcutaneous tissues, an increase in muscle fiber excitability, amplitude of fiber action potentials, and/or conduction velocity due to training-induced increases in Na^+-K^+ pump concentration and/or activity (Hicks, Cupido, Martin, & Dent, 1992; Medbo, Jebens, Vikne, Refnes, & Gramvik, 2001).

It is also important to note the greater modifications observed for VM as compared to VL. A lack of significant increase in iEMG activity for VL associated with ST is similar to a modification reported in an earlier study (Ferri et al., 2003). Pincivero, Salfetnikov, Campy, & Coelho (2004) investigated activation of the quadriceps femoris at different joint angles during MVC. Their results showed better efficiency of the VM recruitment than for the VL muscle between the angles of 70 to 90º. In our study, we evaluated the isometric Cf-t at an angle of 90º, which this may explain the greater magnitude of VM response.

The experimental protocol in our investigation does not allow us to disregard the possibility that adaptations in tendon and muscle structure may have occurred and contributed to the modifications in RFD and MVC. The primary role of tendons is the transmission of contractile forces from muscle to bone. In older adults, the increase in tendon compliance may result in a slower transmission of contractile forces to the skeleton, which would be seen as a slower RFD (Reeves, 2006). Reeves, Narici, and Maganaris (2004) showed that 14 weeks of ST in older adults increased patella tendon stiffness by 64%. These training-induced adaptations have important implications for RFD, suggesting that a stiffer musculotendinous unit would be more effective during the initial transmission of contractile forces and, thus, would increase RFD.

Conclusions

Our findings demonstrate that ST effectively induces marked increases in RFD and EMG amplitudes of the VM in older women; however, the variation in one variable is not correlated significantly with the variation in another. In other words, the rapid muscle force capacity of the neuromuscular system remains trainable in older adults, even during periods as short as 50 ms. This could have important implications for the design of ST programs.

References

Aagaard, P., Simonsen, E. B., Andersen, J. L., Magnusson, P., & Dyhre-Poulsen, P. (2002). Increased rate of force development and neural drive of human skeletal muscle following

resistance training. *Journal of Applied Physiology, 93*, 1318–1326. doi:10.1152/japplphysiol.00283.2002 8750-7587/02

American College of Sports Medicine. (2009). American College of Sports Medicine position stand. Progression models in resistance training for healthy adults. *Medicine & Science in Sports & Exercise, 41*, 687–708. doi:10.1249/MSS.0b013e3181915670

Andersen, L. L., Andersen, J. L., Magnusson, S. P., Suetta, C., Madsen, J. L., Christensen, L. R., & Aagaard, P. (2005). Changes in the human muscle force-velocity relationship in response to resistance training and subsequent detraining. *Journal of Applied Physiology, 99*, 87–94. doi:10.1152/japplphysiol.00091.2005

Barry, B. K., Warman, G. E., & Carson, R. G. (2005). Age–related difference in rapid muscle activation after rate of force development training of the elbow flexors. *Experimental Brain Research, 162*, 122–132. doi:10.1007/s00221-004-2127-3

Benson, C., Docherty, D., & Brandenburg, J. (2006). Acute neuromuscular responses to resistance training performed at different loads. *Journal of Science and Medicine in Sport, 9*, 135–142. doi:10.1016/j.jsams.2005.07.001

Blazevich, A. J., Horne, S., Cannavan, D., Coleman, D. R., & Aagaard, P. (2008). Effect of contraction mode of slow–speed resistance training on the maximum rate of force development in the human quadriceps. *Muscle & Nerve, 38*, 1133–1146. doi:10.1002/mus.21021

Caserotti, P., Aagaard, P., Larsen, B. J., & Puggaard, L. (2008). Explosive heavy-resistance training in old and very old adults: Changes in rapid muscle force, strength and power. *Scandinavian Journal of Medicine & Science in Sports, 18*, 773–782. doi:10.1111/j.1600-0838.2007.00732.x

Christie, A., & Kamen, G. (2006). Doublet discharges in motoneurons of young and older adults. *Journal of Neurophysiology, 95*, 2787–2795. doi:10.1152/jn.00685.2005 0022-3077/06

de Andrade, A. D., Silva, T. N., Vasconcelos, H., Marcelino, M., Rodrigues-Machado, M. G., Filho, V. C.,...Amorim, C. F. (2005). Inspiratory muscular activation during threshold therapy in elderly healthy and patients with COPD. *Journal of Electromyography and Kinesiology, 15*, 631–639. doi:10.1016/j.jelekin.2005.06.002

Ferri, A., Scaglioni, G., Pousson, M., Capodaglio, P., Van Hoecke, J., & Narici, M. V. (2003). Strength and power changes of the human plantar flexors and knee extensors in response to resistance training in old age. *Acta Physiologica Scandinavica, 177*, 69–78. doi:10.1046/j.1365-201X.2003.01050.x

Foldvari, M., Clark, M., Laviolette, L. C., Bernstein, M. A., Kaliton, D., Castaneda, C.,…Singh, M. A. (2000). Association of muscle power with functional status in community-dwelling elderly women. *The Journals of Gerontology. Series A, Biological Sciences and Medical Sciences, 55*, M192–M199. doi:10.1093/gerona/55.4.M192

Garland, S. J., & Griffin, L. (1999). Motor unit double discharges: Statistical anomaly or functional entity? *Canadian Journal of Applied Physiology, 24*, 113–130. Retrieved from http://www.edb.utexas.edu/faculty/griffin/pdf/7-Garland%20&%20Griffin%20(1999).pdf

Godefroy, O., Roussel, M., Despretz, P., Quaglino, V., & Boucart, M. (2010). Age-related slowing: Perceptuomotor, decision, or attention decline? *Experimental Aging Research, 36*, 169–189. doi:10.1080/03610731003613615

Häkkinen, K., Pakarinen, A., Kraemer, W. J., Häkkinen, A., Valkeinen, H., & Alen, M. (2001). Selective muscle hypertrophy, changes in EMG and force, and serum hormones during strength training in older women. *Journal of Applied Physiology, 91*, 569–580. Retrieved from http://jap.physiology.org/cgi/reprint/91/2/569

Henwood, T. R., Riek, S., & Taaffe, D. R. (2008). Strength versus muscle power-specific resistance training in community–dwelling older adults. *The Journals of Gerontology. Series A, Biological Sciences and Medical Sciences, 63*, 83–91. Retrieved from http://biomedgerontology.oxfordjournals.org/content/63/1/83.full

Hermens, H. J., Freriks, B., Disselhorst-Klug, C., & Rau, G. (2000). Development of recommendations for SEMG sensors and sensor placement procedures. *Journal of Electromyography and Kinesiology, 10,* 361–374. doi:10.1016/S1050-6411(00)00027-4

Hicks, A. L., Cupido, C. M., Martin, J., & Dent, J. (1992). Muscle excitation in elderly adults: The effects of training. *Muscle & Nerve, 15*, 87–93. doi:10.1002/mus.880150115

Izquierdo, M., Ibañez, J., Gorostiaga, E., Garrues, M., Zúñiga, A., Antón, A.,...Häkkinen, K. (1999). Maximal strength and power characteristics in isometric and dynamic actions of the upper and lower extremities in middle-aged and older men. *Acta Physiologica Scandinavica, 167*, 57–68. doi:10.1046/j.1365-201x.1999.00590.x

Kalapotharakos, V. I., Smilios, I., Parlavatzas, A., & Tokmakidis, S. P. (2007). The effect of moderate resistance strength training and detraining on muscle strength and power in older men. *Journal of Geriatric Physical Therapy, 30*, 109–113. Retrieved from http://journals.lww.com/jgpt/Abstract/2007/12000/The_Effect_of_Moderate_Resistance_Strength.5.aspx

Kamen, G. (2005). Aging, resistance training, and motor unit discharge behavior. *Canadian Journal of Applied Physiology, 30*, 341–51.

Kamen, G., & Knight, C. A. (2004). Training-related adaptations in motor unit discharge rate in young and older adults. *The Journals of Gerontology. Series A, Biological Sciences and Medical Sciences, 59*, 1334–1338. doi:10.1093/gerona/59.12.1334

Katula, J. A., Rejeski, W. J., & Marsh, A. P. (2008). Enhancing quality of life in older adults: A comparison of muscular strength and power training. *Health and Quality of Life Outcomes, 6*, 45. doi:10.1186/1477-7525-6-45

Klass, M., Baudry, S., & Duchateau, J. (2008). Age-related decline in rate of torque development is accompanied by lower maximal motor unit discharge frequency during fast contractions. *Journal of Applied Physiology, 104*, 739–746. doi:10.1152/japplphysiol.00550.2007

Kraemer, W. J., & Newton, R. U. (2000). Training for muscular power. *Physical Medicine and Rehabilitation Clinics of North America, 11*, 341–368.

Kwon, I. S., Schrager, M., Talbot, L. A., Fozard, J. L., & Metter, E. J. (2001). Relationship between muscle strength and the time taken to complete a standardized walk-turn-walk test. *The Journals of Gerontology. Series A, Biological Sciences and Medical Sciences, 56*, B398–40. doi:10.1093/gerona/56.9.B398

Latham, N. K., Bennett, D. A., Stretton, C. M., & Anderson, C. S. (2004). Systematic review of progressive resistance strength training in older adults. *The Journals of Gerontology,*

Series A, Biological Sciences and Medical Sciences, *59*, 48–61. doi:10.1093/gerona/59.1.M48

Li, Y., McColgin, C., & Van Oteghen, S. L. (1998). Comparisons of psychomotor performance between the upper and lower extremities in three age groups. *Perceptual and Motor Skills*, *87*, 947–52. doi:10.2466/PMS.87.7.947-952

Lynch, N. A., Metter, E. J., Lindle, R. S., Fozard, J. L., Tobin, J. D., Roy, T. A.,…Hurley, B. F. (1999). Muscle quality. I. Age-associated differences between arm and leg muscle groups. *Journal of Applied Physiology*, *86*, 188–194. Retrieved from http://jap.physiology.org/cgi/content/full/86/1/188

Marmeleira, J. F., Godinho, M. B., & Fernandes, O. M. (2009). The effects of an exercise program on several abilities associated with driving performance in older adults. *Accident; Analysis and Prevention*, *41*, 90–97. doi:10.1016/j.aap.2008.09.008

Medbo, J., Jebens, E., Vikne, H., Refnes, P., & Gramvik, P. (2001). Effects of strenuous strength training on the Na-K pump concentration in skeletal muscle of well-trained men. *European Journal of Applied Physiology*, *84*, 148–154. doi:10.1007/s004210000311

Merletti, R. (1999). Standards for reporting EMG data (ISEK/1999). *Journal of Electromyography and Kinesiology*, *9*, III–IV. Retrieved from http://www.isek-online.org/standards_emg.html

Pijnappels, M., van der Burg, P. J., Reeves, N. D., & van Dieën, J. H. (2008). Identification of elderly fallers by muscle strength measures. *European Journal of Applied Physiology*, *102*, 585–592. doi:10.1007/s00421-007-0613-6

Pincivero, D. M., Salfetnikov, Y., Campy, R. M., & Coelho, A. J. (2004). Angle- and gender-specific quadriceps femoris muscle recruitment and knee extensor torque. *Journal of Biomechanics*, *37*, 1689–1697. doi:10.1016/j.jbiomech.2004.02.005

Reeves, N. D. (2006). Adaptation of the tendon to mechanical usage. *Journal of Musculoskeletal & Neuronal Interactions*, *6*, 174–180. Retrieved from http://hdl.handle.net/2173/6686

Reeves, N. D., Narici, M. V., & Maganaris, C. N. (2004). In vivo human muscle structure and function: Adaptations to resistance training in old age. *Experimental Physiology*, *89*, 675–689. doi:10.1113/expphysiol.2004.027797

Sahaly, R., Vandewalle, H., Driss, T., & Monod, H. (2001). Maximal voluntary force and rate of force development in humans: Importance of instruction. *European Journal of Applied Physiology*, *85*, 345–350. doi:10.1007/s004210100451

Skelton, D. A., Greig, C. A., Davies, J. M., & Young, A. (1994). Strength, power and related functional ability of healthy people aged 65–89 years. *Age and Ageing*, *23*, 371–377. doi:10.1093/ageing/23.5.371

Skelton, D. A., Kennedy, J., & Rutherford, O. M. (2002). Explosive power and asymmetry in leg muscle function in frequent fallers and non-fallers aged over 65. *Age and Ageing*, *31*, 119–125. doi:10.1093/ageing/31.2.119

Suetta, C., Aagaard, P., Rosted, A., Jakobsen, A. K., Duus, B., Kjaer, M., & Magnusson, S. P. (2004). Training-induced changes in muscle CSA, muscle strength, EMG, and rate of force development in elderly subjects after long-term unilateral disuse. *Journal of Applied Physiology*, *97*, 1954–1961. doi:10.1152/japplphysiol.01307.2003

Van Cutsem, M., Duchateau, J., & Hainaut, K. (1998). Changes in single motor unit behavior contribute to the increase in contraction speed after dynamic training in humans. *Journal of Physiology*, *513*, 295–305. doi:10.1111/j.1469-7793.1998.295by.x

Authors' Notes

This research was supported by: Financiadora de Estudos e Projetos (FINEP); Fundação para o Desenvolvimento da UNESP (FUNDUNESP) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq).
Please address correspondence concerning this article to Sebastião Gobbi, Departamento de Educação Física, UNESP, Avenida 24 A, 1515, Bela Vista, 13506-900, Rio Claro – SP, Brazil.

E-mail: sgobbi@rc.unesp.br